TOBIAS I. LIAUDAT

Ph.D. in Physics

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Nationality: Argentinian *Last update: October 27, 2026*

PROFESSIONAL EXPERIENCE

IRFU, CEA Saclay Dec. 2023 - Now
Research Scientist *Saclay, France*

· Staff research scientist at the LILAS laboratory of the DEDIP department in the Institute of research into the fundamental laws of the Universe (IRFU).

University College London, Computer Science Department Dec. 2022 - Nov. 2023
Research Fellow in Artificial Intelligence for Imaging *London, UK*

· Research fellowship of 2 years to work with Profs Jason McEwen, Marcelo Pereyra and Marta Betcke.

École Polytechnique, Applied Mathematics Department Aug. 2020 - Aug. 2022
Teaching assistant *Palaiseau, France*

· Teaching for bachelor and MSc. students in the top ranked French engineering school.

École Nationale de la Statistique de l'Administration Économique Oct. 2018 - Feb. 2019
Research assistant *Palaiseau, France*

· Topic: Regularized optimal transport for signed measures. Supervisor: Marco Cuturi.

CosmoStat, CEA Saclay Mar. 2018 - Sep. 2018
Research internship *Saclay, France*

· Topic: Distributed sparse blind source separation for very large-scale datasets. Supervisor: Jérôme Bobin.

Thales Air Systems Jul. 2017 - Sep. 2017
Summer internship *Limours, France*

· Topic: Development of tools to evaluate civil radar performances. Supervisor: Daniel Nguyen.

EDUCATION

CEA Saclay / Université Paris-Saclay Nov. 2019 - Oct. 2022
Ph.D. in Physics *Saclay, France*

· Topic: Data-driven modelling of ground-based and space-based telescope's point spread functions.
· Supervisors: Jean-Luc Starck & Martin Kilbinger.

Université de Rennes I / CentraleSupélec / IMT Atlantique Aug. 2017 - Sep. 2018
Master 2 Research - SISEA *Brest, France*

· Specialization in signal and image processing. Joint with IMT Atlantique engineering degree.
· Ranked 1st out of all master students.

IMT Atlantique (*Télécom Bretagne*) Aug. 2016 - Sep. 2018
Telecommunications Engineering degree *Brest, France*

· One of France's top-engineering schools. Specialization in signal and image processing, and machine learning.
· Recipient of the Eiffel excellence scholarship for a joint degree with the University of Buenos Aires.

Engineering School of the University of Buenos Aires Mar. 2012 - Sep. 2019
Electronic Engineering degree *Buenos Aires, Argentina*

· Specialization in signal and image processing, electronic circuits and computer science. GPA: 9.23/10.
· Ranked 4th of 1634 students when starting the joint degree in 2016.

MEMBERSHIP IN SCIENTIFIC COLLABORATIONS

LISA consortium	Dec 2023 - Present
Euclid consortium	Nov. 2019 - Present
UNIONS/CFIS collaboration	Nov. 2019 - Present
COSMOS-Webb collaboration	May 2022 - Present

TECHNICAL SKILLS

Main programming language	Python (JAX, PyTorch, TensorFlow, ...)
Programming languages I have used	MATLAB, C, Java, Shell, Assembly
Other tools	Git, LaTex, HPC (SLURM, TORQUE, SMP, MPI), CI, PyPI Experience with CPU and GPU computer clusters
Astronomical software	PSFEx, SExtractor, GalSim, ShapePipe

STUDENT SUPERVISION

Postdoc supervision

- Tom Sprunck, (2025-2026). Topic: Machine learning for scientific imaging in astrophysics.
  Now *chargé de recherche* at CEREMA, Strasbourg, France.

PhD student cosupervision

- Sammy Sharief, (2025-Present). Topic: Generative AI for Robust Uncertainty Quantification in Astrophysical Inverse Problems.
- Louis Le Saulnier, (2025-Present). Topic: Fast massive black hole binaries parameter estimation for LISA.
- Henry Aldridge, (2023-Present). Topic: Uncertainty quantification and model comparison in high dimensions. Main supervisor: Prof. Jason McEwen (UCL).
- Jessica Whitney, (2022-Present). Topic: Machine learning approaches for dark matter mass mapping. Main supervisor: Prof. Jason McEwen (UCL).

Master student supervision

- Seif Zaafouri, 6 months (2025-2026). Topic: Conformal prediction for high-dimensional data.
- Louis Le Saulnier, 6 months (2025). Topic: Fast massive black hole binaries parameter estimation for LISA.
- Khaoula Arouissi, 6 months (2025). Topic: Equivariant boostrap for radio interferometry.
- Mostafa Cherif, 6 months (2024). Topic: Uncertainty quantification for radio interferometry.
- Ezequiel Centofanti, 6 months (2022). Topic: Improving WaveDiff PSF estimation for *Euclid.*
- Aziz Ayed, 5 months (2021). Topic: Deep denoisers for the MCCD PSF model.
- Jérôme Bonnin, 6 months (2019-2020). Topic: RCA for CFIS and the MCCD PSF model.

SELECTED PRESENTATIONS AND CONFERENCES

Presentations within the Euclid, LISA Consortiums are omitted.

Contributed poster at ICML 2026.	Seoul, South Korea. Jul 2026
Invited talk at the Lumière conference.	Institut Pascal, Saclay, France. Jan 2026
Invited talk at Conference Inverse Problems in Imaging, Foundations & applications.	CIRM, France. Oct 2025
Invited seminar at Data science seminar.	CEA Cadarache, France. Sep 2025
Invited seminar at CEA Saclay on AI for science.	Saclay, France. Jun 2025
Invited seminar at the Scattering club of the LPENS, ENS Ulm.	Paris, France. Apr. 2025
Invited seminar at La Maison de la Simulation.	CEA Saclay, France. Apr. 2025
Invited talk at NumPEx general assemby.	INRIA Saclay, France. Mar. 2025

Contributed poster at the BASP Frontiers 2025 conference. Villars-sur-Ollon, Switzerland. Jan. 2025
Invited talk at the Deep CosmoStat Days workshop. Saclay, France. Jan. 2025
Invited seminar at the InTheArt seminar series. Saclay, France. Nov. 2024
Invited talk at the Seminaire de la Mission Numérique meeting. Grenoble, France. Nov. 2024
Invited talk at the PEPR NumPEx Exa-DI workshop. Paris, France. Oct. 2024
Contributed talk at the Cosmo21 conference. Chania, Greece. May. 2024
Invited seminar at the Data Analysis Group at IRFU, CEA Saclay. Saclay, France. March. 2024
Invited speaker at the Applied Inverse Problems (AIP) conference. Göttingen, Germany. Sep. 2023
Contributed talk at the European Astronomical Society meeting (EAS 2023). Krakow, Poland. Jul. 2023
Invited seminar at ULC Computer Science Department CMIC. London, UK. Jun. 2023
Invited seminar at the Imaging in Paris seminar. Insitut Henri Poincaré, Paris. Jun. 2023
Invited speaker at the BASP Frontiers conference. Switzerland, Feb. 2023
Invited poster at Interfacing Bayesian statistics, machine learning, applied analysis, and blind and semi-blind imaging inverse problems' workshop. Edinburgh, Jan. 2023
Invited seminar at the School of Math. and Computer Sciences at Heriot-Watt Univ. Edinburgh, Sep. 2022
Invited seminar to the BASP group at Heriot-Watt Univ. Online. Mar. 2022
Invited seminar to the PSF modelling group of the LSST DESC collaboration. Online. Mar. 2022
Talk at the SIAM 2022 Conference on Imaging Science (IS22). Online. Mar. 2022
Poster at the 2021 NeurIPS, Machine Learning and the Physical Sciences Workshop. Online. Dec. 2021
Talk at the lancement de l'axe astrophysique de la Graduate School Physique. Saclay, France. Nov. 2021
Talk at the 52èmes Journées de Statistiques de la Société Française de Statistique. Online. Jun. 2021
Talk at the Peyresq summer school on signal and image processing. Online. Jun. 2021
Talk at the 2021 UNIONS CFIS/Pan-STARRS/WISHES collaboration meeting. Online. Mar. 2021
Poster at the SPARS conference. Toulouse, France. Jul. 2019
Invited seminar at CEA CosmoStat's group. Saclay, France. Feb. 2019

TEACHING

***Qualification* section 61 CNU, France** Feb. 2023

- *Qualification* to be able to apply for professor positions in France's universities. CNU section 61 corresponds to computer science, control engineering and signal processing.

École Polytechnique, Applied Mathematics Department Aug. 2020 - Aug. 2022

Teaching assistant at France's top engineering school. MSc and bachelor students.

Year 2021-2022 (32h)

- *Optimization and control* [MAP435] for MSc. students with Prof. Grégoire Allaire.
- *Applied mathematics python projects* [MAP361P] for MSc. students with Prof. Arvind Singh.

Year 2020-2021 (64h)

- *Statistics* [MAP433] for MSc. students with Prof. Eric Moulines.
- *Optimization and control* [MAP435] for MSc. students with Prof. Grégoire Allaire.
- *Mathematical modelling* [MAA107] for Bac. students with Profs. Vincent Bansaye and Thibaut Mastrolia.
- *Applied mathematics python projects* [MAP361P] for MSc. students with Prof. Arvind Singh.

AWARDS & DISTINCTIONS

Distinguished student award University of Buenos Aires, Argentina, 2017
Eiffel excellence scholarship Campus France, 2016
- Awarded to outstanding international students to do a joint degree in France.

Gold medal award St. Luke's College, Buenos Aires, Argentina, 2011
- Delivered to the best student of the 2011 class.

MISCELLANEOUS

- Fluent in Spanish (mother tongue), English and French.

- Co-organizer of the Seminaire Palaisien and previous organizer of the CosmoStat Journal club.
- Reviewed articles for ApJ, MNRAS, NeurIPS, Astronomy & Astrophysics (A&A),The Astronomical Journal,RAS Techniques & Instruments (RASTI), Journal of Open Source Software (JOSS), IEEE Transactions on Computational Imaging (TCI),Philosophical Transactions of the Royal Society A (RSTA).